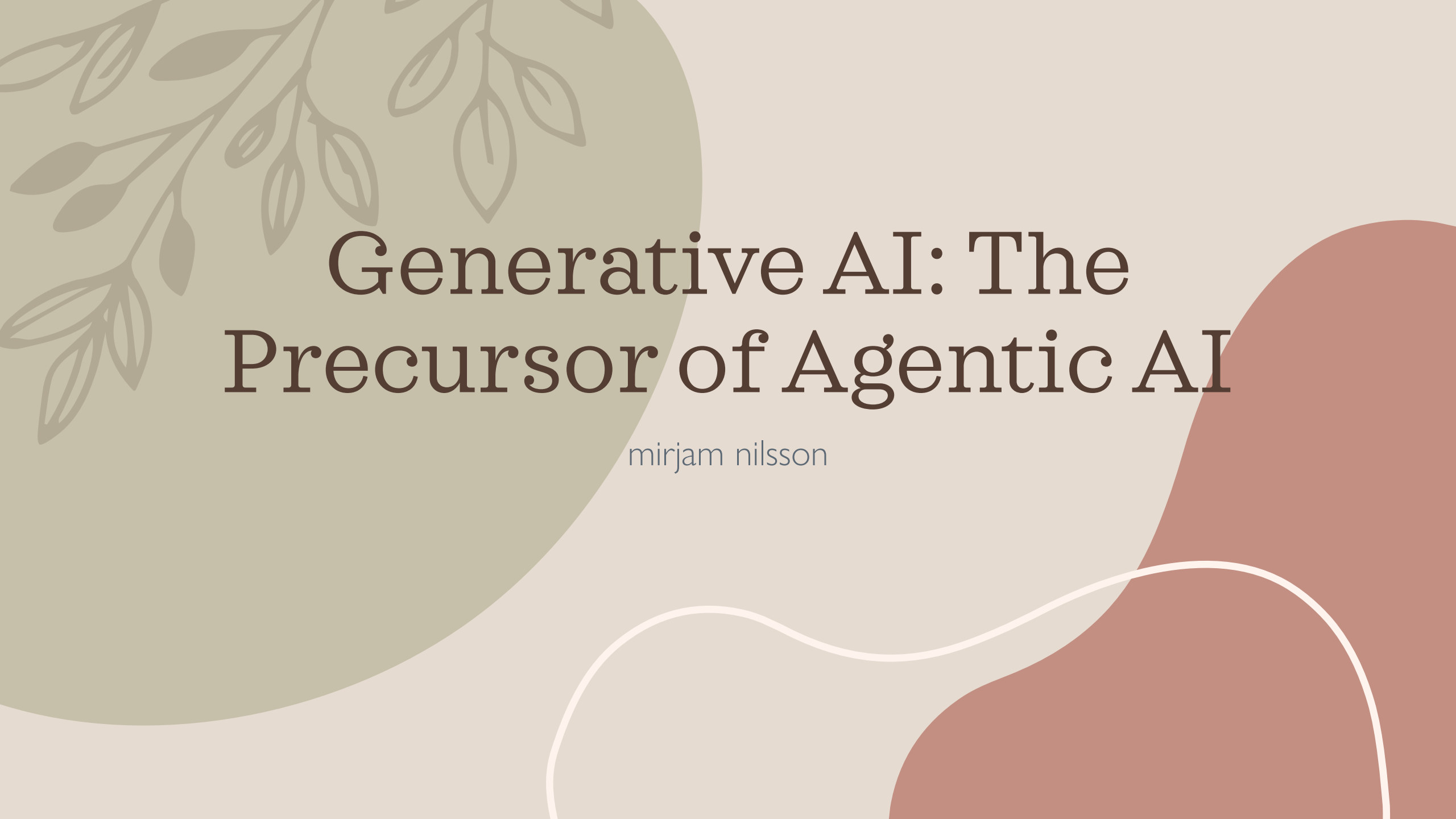




Generative AI: The Precursor of Agentic AI

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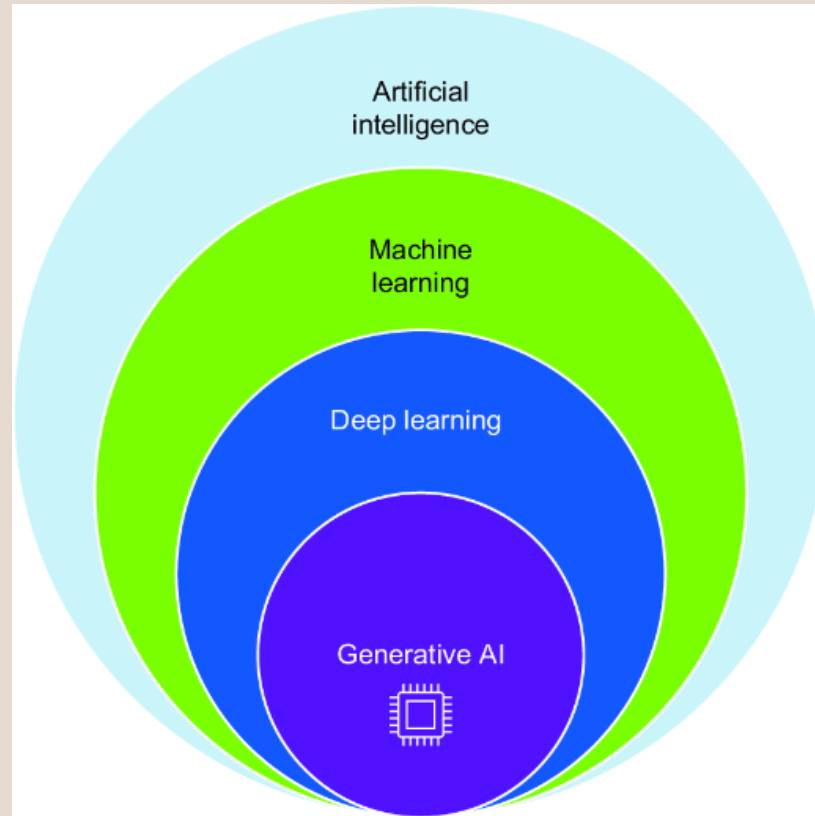
What is Generative AI?

- Generative AI, refers to a category of artificial intelligence models that are capable of generating new (novel) and original content.
- This content can take various forms, such as images, text, code, or even entire datasets.
- Generative AI systems have the capability to produce content that was not explicitly programmed or provided in the training data.

Generative AI is a Deep Learning Technology

- Generative AI is strongly based on deep learning models and word embedding technologies
- As we are going to see, deep learning models succeeded in capturing the hidden rules for generating or understanding language with surprising details
- While human beings focus on upper grammar, syntax, lexical and semantic rules, deep learning models discover and capture micro-level grammar, syntax and semantic rules well-welded in low level math structures
- The capabilities of deep learning models have been greatly boosted by word embedding technologies that made it possible to learn low level contexts of words and sentences

Generative AI is a subset of Deep Learning



Main Goal

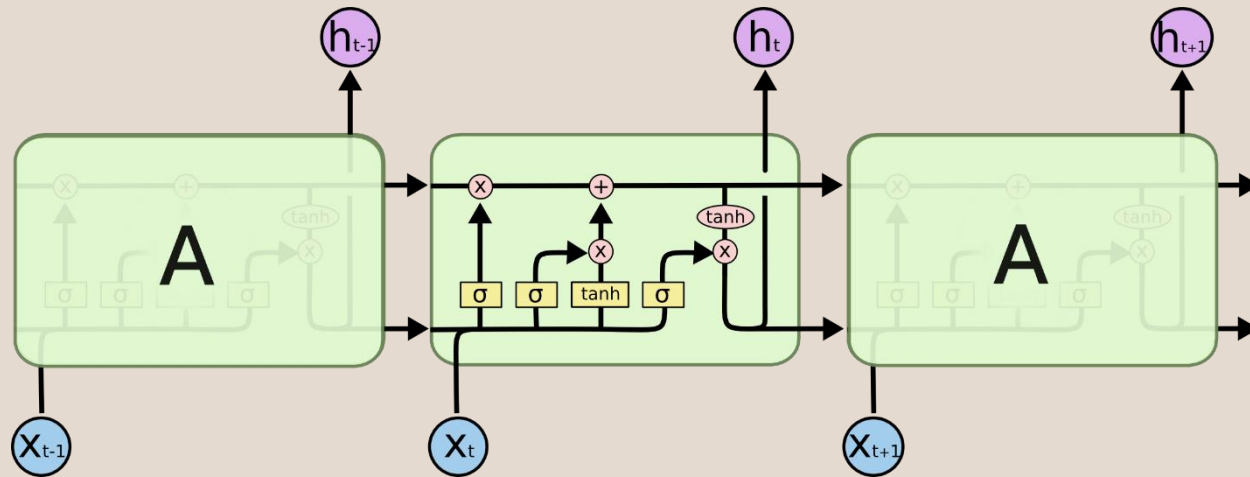
- The primary goal of generative models is to understand and learn patterns from existing data so well that they can generate similar, but new, data points.

Large Language Models: LLM's

- In the early days of AI, deep learning models were able to handle only small datasets due to limitations of hardware and the lack of data
- In the Big Data era deep learning models are now able to handle huge datasets, which led to what we call Large Language Models (LLM's)

Sequential LSTM Models

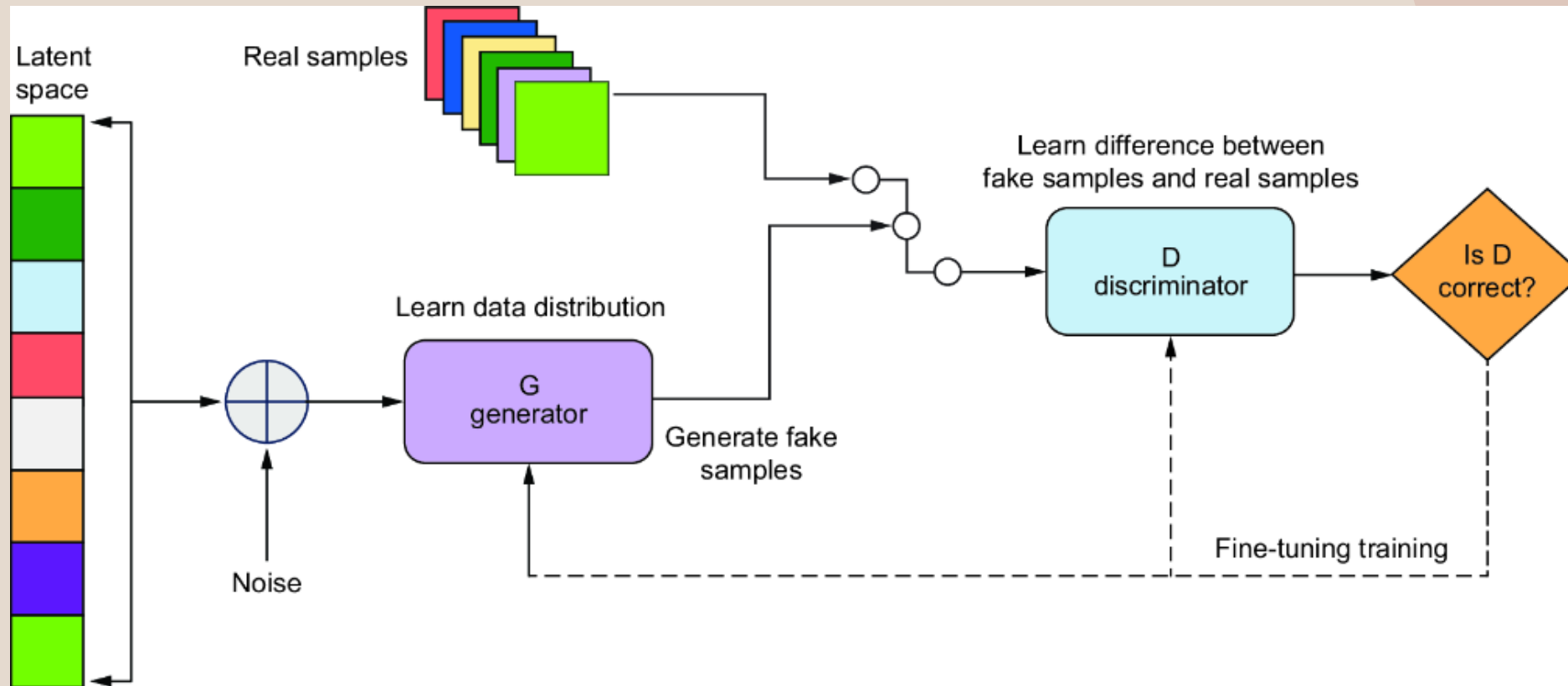
- **Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM):** These are types of neural networks that can be used for sequence generation tasks, such as generating sentences, paragraphs, or even entire articles.



Generative Adversarial Models

- **Generative Adversarial Networks (GANs):** GANs consist of two neural networks, a generator, and a discriminator, which are trained simultaneously through adversarial training.
- The generator creates data, and the discriminator evaluates how closely it resembles real data.
- The two networks improve iteratively, with the generator aiming to generate realistic content and the discriminator becoming better at distinguishing between real and generated data

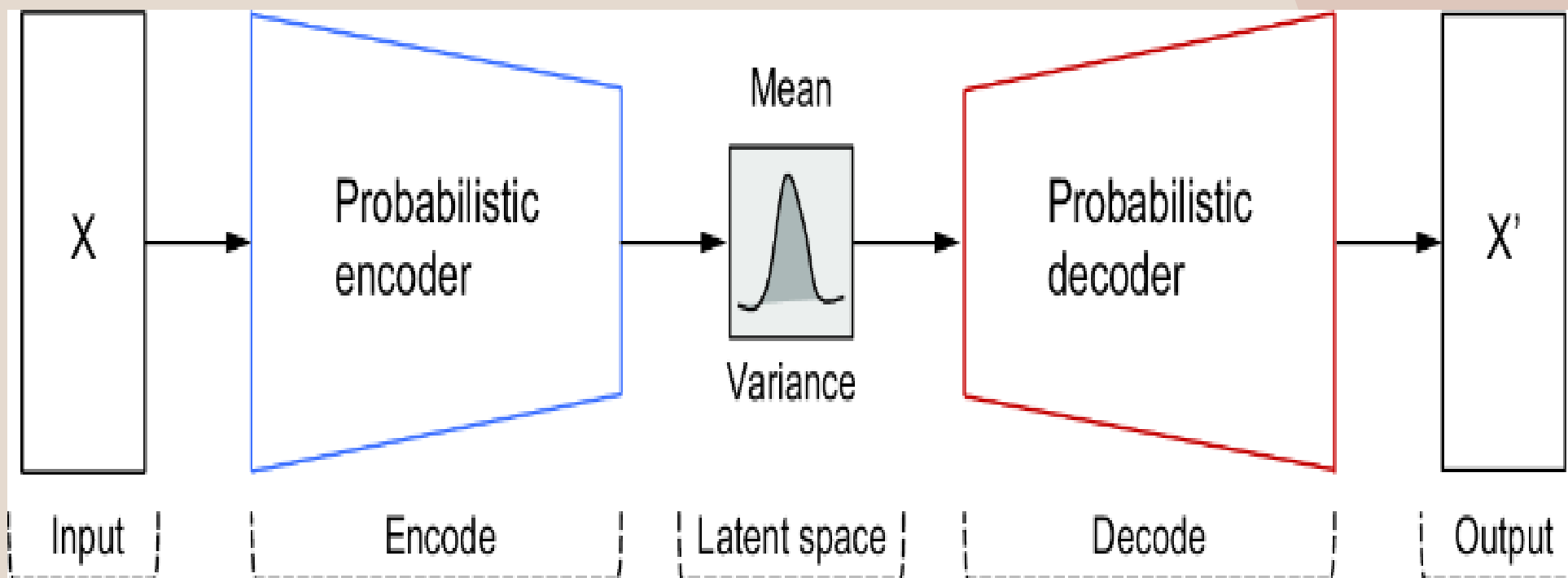
GAN



Variational Autoencoders (VAEs)

- **Variational Autoencoders (VAEs):** VAEs are a type of generative model that learns a probabilistic mapping between input data and latent space.
- This latent space is then used to generate new data points that share similar characteristics with the training data

VAE's



GPT's

- GPT, short for "**Generative Pre-trained Transformer**," is a family of natural language processing models developed by OpenAI.
- GPT models are designed to understand and generate human-like text, making them versatile for various language-related tasks, including text generation, translation, summarization, question-answering, and more.
- These models are built on the Transformer architecture, which has become a cornerstone in modern NLP.

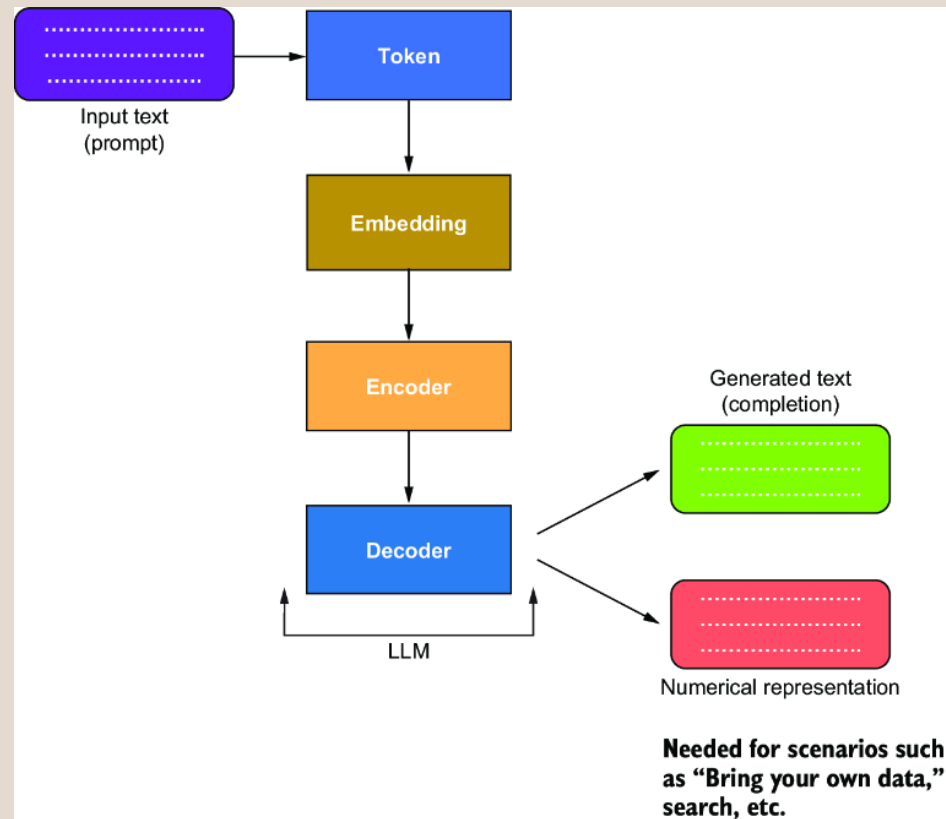
What is ChatGPT

- ChatGPT, is a model developed by OpenAI, is an advanced AI conversational agent. It is an instance of the GPT-4 model, which is a part of the Generative Pretrained Transformer series.
- At its core, ChatGPT uses machine learning techniques to generate human-like text based on the input it receives. It can understand the context of a conversation, answer questions, provide explanations, and even generate creative content, such as stories or poems.
- The "GPT" in ChatGPT stands for "Generative Pretrained Transformer," which represents the foundational technology that powers it:
 - **Generative** signifies the model's capability to generate text rather than just understanding or classifying it.
 - **Pretrained** indicates that the model has undergone an initial training phase where it learned from a large corpus of Internet text.
 - **Transformer** refers to the type of model architecture used, which helps understand the context of the input text.
 - This combination of technologies enables ChatGPT to generate meaningful responses which closely mimic natural human conversation.

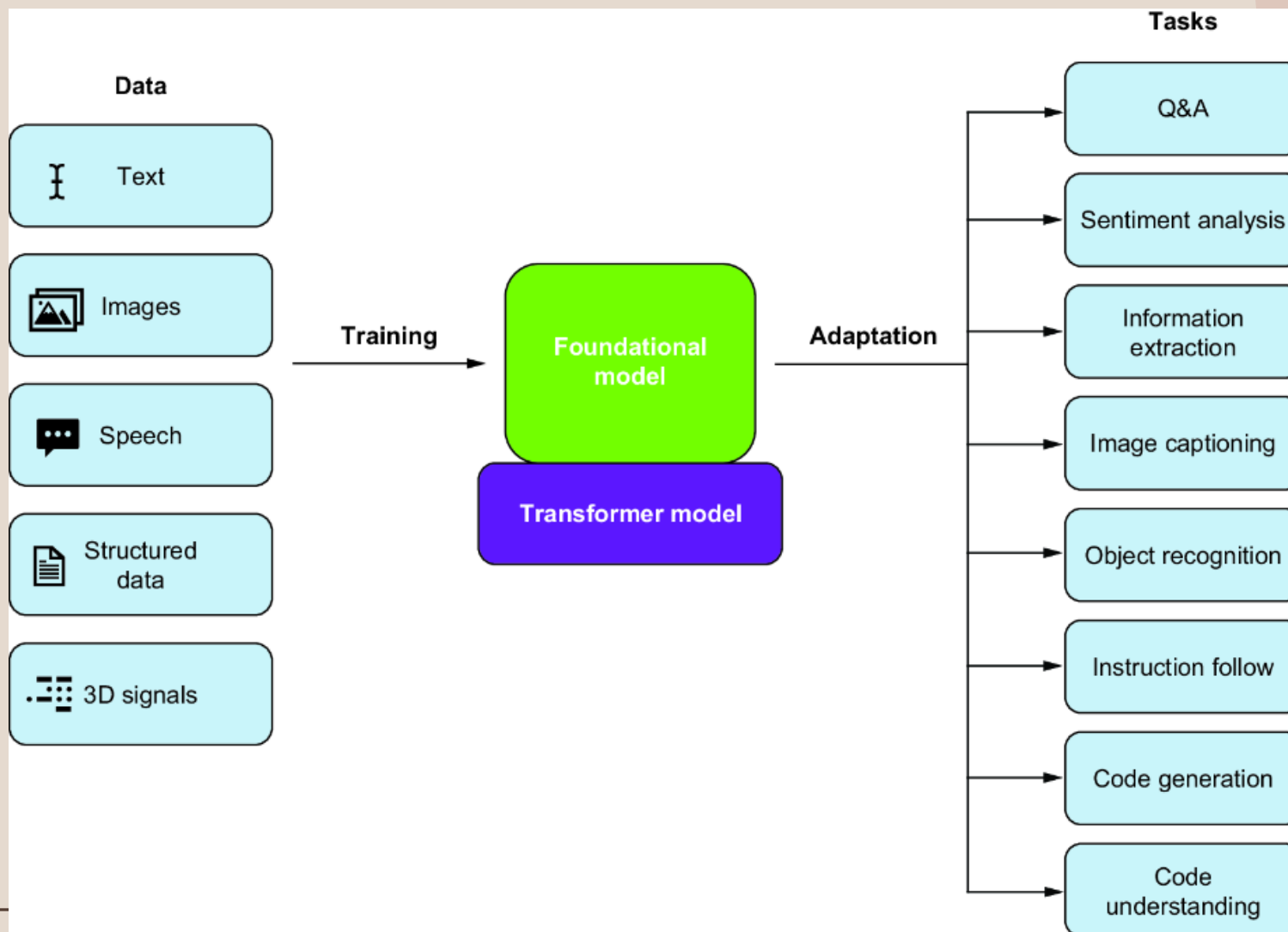
Transformer Models

- **Transformer Models:** Transformer models, like GPT (Generative Pre-trained Transformer), have gained popularity for their ability to generate coherent and contextually relevant text. GPT, for instance, is pre-trained on vast amounts of text data and can then be fine-tuned for specific generative tasks

The Transformer



LLM and the Transformer



The Importance and Relevance of ChatGPT

- For many years humans have been trying to interact with computers using natural language instead of programming languages.
- The dreams include writing applications, asking computers to build products, consultations, answering questions, solving problems, generating content.
- ChatGPT is a dream-came-true
- Although released only recently, its ability to understand and generate human-like text is surprising and has a seemingly endless range of applications.
- It's a valuable tool in areas such as customer service and communication, content generation, education, research, language learning, data science, and many more.

ChatGPT Pretraining

- ChatGPT is a Large Language Model (LLM) built upon a transformer architecture and is trained using a variant of unsupervised learning known as self-supervised learning.
- This approach is a bit different from traditional unsupervised learning methods, but it's often categorized under the same umbrella due to its lack of reliance on labeled data.

Self-Supervised Learning

- 1. Concept:** Self-supervised learning is a type of training where the model generates its own labels from the input data. For language models like ChatGPT, this typically involves predicting parts of the text given other parts.
- 2. Data Processing:** The model is fed with large amounts of text data.
3. This data can include books, articles, websites, and other written material.
4. The text is tokenized (converted into a sequence of tokens, such as words or subwords) before being fed into the model.

Training Algorithm - Next Word Prediction

- **Masked Language Modeling (MLM):** In models like BERT, some tokens in a sequence are randomly masked, and the model is trained to predict these masked tokens based only on their context.
- **Autoregressive Modeling:** In models like GPT (which includes ChatGPT), the training involves predicting the next word in a sequence given the previous words. This is an autoregressive process, as the prediction for each word depends on the words that came before it.

Pretraining and Fine-Tuning

- **Pre-training:** GPT models are initially pre-trained on massive text corpora from the internet. During this phase, the model learns to predict the next word in a sentence given the preceding context. It builds a strong understanding of grammar, vocabulary, and world knowledge.
- **Transformer Architecture:** GPT models use the Transformer architecture, which relies on self-attention mechanisms to capture contextual information from input sequences effectively. This allows the model to handle long-range dependencies and understand the context of a word or phrase within a sentence.
- **Fine-tuning:** After pre-training, GPT models can be fine-tuned on specific tasks or datasets. For example, they can be fine-tuned for language translation by providing pairs of sentences in different languages. Fine-tuning helps adapt the model's general language understanding to more specific tasks.

Interacting With ChatGPT: **Prompts**

- A "prompt" in the context of ChatGPT or other GPT models is the input text that you provide to the model.
- This input text guides the model's response.
- The design of these prompts is crucial because the model's output heavily depends on the quality and structure of the input it receives.
- Prompts can range from simple questions to complex instructions, and the model generates responses based on its training on a vast corpus of text data.
- The effectiveness of a prompt often determines how relevant, accurate, or creative the model's output will be.

The Art and Science of **Prompt Engineering**

Prompt engineering is the skillful crafting of input prompts to elicit the desired response from a language model.

It's a blend of art and science, requiring an understanding of the model's capabilities and limitations, as well as creativity in phrasing prompts.

Example prompts and applications

1.Direct Question:

1. Prompt: "What is the capital of Finland?"

2.Instruction-Based:

1. Prompt: "Write a summary of the novel '1984' by George Orwell."

3.Creative Generation:

1. Prompt: "Compose a poem about the ocean in the style of Shakespeare."

Example Prompts and Applications?

1.Problem Solving:

1. Prompt: "How can I fix a leaking faucet?"

2.Contextual Conversation:

1. Prompt: "I'm feeling stressed about course load this semester. Do you have any advice?"

3.Coding Help:

1. Prompt: "Write a Python function to calculate the factorial of a number."

Tuning

1. Supervised Fine-Tuning:

1. **Process:** This involves training the model on a dataset of human-generated responses. The dataset typically consists of prompts and the corresponding ideal responses.
2. **Effectiveness:** Highly effective for improving response quality in specific domains or use cases. It helps the model learn context-specific nuances.

2. Reinforcement Learning from Human Feedback (RLHF):

1. **Process:**
 1. **Reward Modeling:** Human trainers rank responses or provide scores to teach the model what constitutes a good response.
 2. **Proximal Policy Optimization (PPO):** The model is updated to generate responses that are more likely to receive high scores.
2. **Effectiveness:** Very effective for aligning the model's responses with human values and preferences. It enhances the model's ability to generate safe, relevant, and contextually appropriate responses.

3. Prompt Engineering and Few-Shot Learning:

1. **Process:** Instead of fine-tuning the model, the input (prompt) is engineered to guide the model's response. Few-shot learning involves providing examples in the prompt to demonstrate the desired response format or content.
2. **Effectiveness:** This is more of a usage technique than a tuning method. It can be highly effective when specific fine-tuning is not feasible. The effectiveness depends on the skill in crafting prompts.

Domain Tuning

1. Domain-Specific Tuning:

1. **Process:** The model is fine-tuned on a specific domain or subject matter, such as legal texts, medical information, or technical documentation.
2. **Effectiveness:** Highly effective for improving accuracy in specialized areas. However, it requires a carefully curated dataset from the specific domain.

2. Hyperparameter Tuning:

1. **Process:** Adjusting the model's hyperparameters (such as learning rate, batch size, number of layers) to optimize performance.
2. **Effectiveness:** Can be effective for improving model training efficiency and stability, but requires deep expertise and experimentation.